



GENERAL MATHEMATICS

Grade 11

**Three-Term Budget Of Work (BOW)
for Learning Competencies**

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Three-Term Budget Of Work For Learning Competencies

First Term	
Content Standard	The learners should have knowledge and understanding of... • business and finance through problems involving earning money, buying, and selling • patterns, arithmetic and geometric sequences and series • financial application of sequence and series • measurement and conversion • functions and its graphs
Performance Standard	By the end of the term, the learners are able to... • Prepare accurate payrolls and digital payslips with correct gross pay, deductions, and net income. • Create and present a digital financial report with graphs, equations, and justified real-life decisions. • Design and present a practical measurement or conversion solution with a concrete or visual model.
Week	Learning Competency
1	Number and Algebra • Calculate weekly or monthly wages from an annual salary, wages from an hourly rate, including situations involving overtime and other allowances, and earnings based on commission or piecework. • Apply a percentage increase or decrease in various contexts: a. determining the impact of inflation on costs and wages over time

	b. calculating percentage mark-ups and discounts c. calculating VAT d. calculating profit or loss in both absolute and percentage terms
2	Number and Algebra • Solve problems involving salaries, wages, benefits, and deductions (tax computations, overtime pay, and gross and net incomes) using appropriate technology. • Solve problems involving salaries, wages, benefits, and deductions (tax computations, overtime pay, and gross and net incomes) using appropriate technology.
3	Number and Algebra • Describe patterns by inspection, including patterns in art and nature. • Determine the next term of a given pattern, including the Fibonacci sequence. • Identify the rule of a given pattern. • Illustrate the attributes of arithmetic and geometric sequences. • Solve problems involving arithmetic and geometric sequences.
4	Number and Algebra • Differentiate sequences from series. • Illustrate arithmetic and geometric series. • Represent series using sigma notation and vice versa. • Find the sum of arithmetic and geometric series. • Solve problems involving arithmetic and geometric series.
5	Number and Algebra • Apply sequences and series to solve financial problems, such as periodic payments for loans and mortgages. • Determine the accumulated values of investment over time using sequences and series.
6	Measurement and Geometry • Determine and apply the appropriate metric units for various contexts, such as units for small and

	large quantities and scientific measurements. • Convert measurements with different units in the same system and between different systems.
7	Measurement and Geometry • Determine the surface area of three-dimensional objects, such as prisms, pyramids, cylinders, cones, spheres, and combinations of these. • Calculate the volume of prisms, pyramids, cylinders, cones, spheres, and combinations of these.
8	Measurement and Geometry • Use a scale to interpret and compare distances on maps, blueprints, and sketches. • Solve problems involving the conversion of currencies. • Apply time conversions in various real-world scenarios such as travel planning, project management, and daily routine schedules for personal and business transactions.
9	Measurement and Geometry • Apply temperature conversions to real-world scenarios such as recipe adjustments, weather reports, scientific experiments, storage conditions, and energy efficiency.
	Number and Algebra • Classify functions based on their graphs and/or equations to determine if they are linear, absolute value, quadratic, square root, or cube root.
10	Number and Algebra • Sketch the graph of the following functions: linear, absolute value, quadratic, square root, or cube root with or without technology. • Apply the functions to model and solve real-life problems.

Second Term	
Content Standard	The learners should have knowledge and understanding of... • piecewise functions • statistical variables • basic trigonometry • practical applications of measurements and transformational geometry • random variables and sampling
Performance Standard	By the end of the term, the learners are able to... • model and represent real-world situations using piecewise functions, and solve practical problems while explaining the reasonableness of results. • Collect, analyze, and interpret data to make decisions using spreadsheets, graphs, and statistical visualizations. • Apply trigonometry and geometry to design and construct 3D objects, showing models or prototypes. • Calculate and analyze random variables, normal distributions, and present data with graphs and tables.
Week	Learning Competency
1	Number and Algebra • Illustrate a piecewise function in practical contexts (i.e., fare rates, purchasing, and income tax). • Solve practical problems involving piecewise functions.
2	Data and Probability • Identify different types of data. • Classify a given set of data according to its level of measurement. • Identify the appropriate measure of central tendency and measures of variability of a given data set for different purposes.

	• Use appropriate technologies (e.g., calculator, Excel, Jamovi, Geogebra, etc.) to calculate measures of central tendency and variability. • Interpret the computed measures of central tendency and variability.
3 to 4	Measurement and Geometry • Illustrate the trigonometric ratios in a right-angled triangle using a real-life object or figure. • Illustrate the area of oblique triangles (i.e., Heron's Formula and the rule $\text{Area} = \frac{1}{2} ab \sin C$). • Solve practical problems involving: a. right and oblique triangles b. angles of elevation and depression c. the use of bearings in navigation
5	Measurement and Geometry • Estimate and measure lengths and distances (e.g., floor dimensions, measuring fabric for clothing). • Calculate the area and perimeter of residential spaces (e.g., room renovations, garden layouts).
6	Measurement and Geometry • Calculate the volume and capacity of three-dimensional objects (e.g., water tanks, rice storage containers, balik-bayan boxes). • Apply volume and capacity in cost estimation (e.g., amount of flour needed for bread production, paint needed for walls).
7	Data and Probability • Differentiate between discrete and continuous random variables and describe their uses. • Calculate the expected value, variance, and standard deviation of discrete random variables.
8	Data and Probability • Illustrate a normal random variable and its characteristics. • Convert a normal random variable to a standard normal variable and vice versa.
9	Data and Probability

	• Compute probabilities and percentiles involving the standard normal distribution using a Z-distribution table or appropriate technologies. • Solve problems involving normal distribution.
10	Data and Probability • Illustrate random sampling. • Determine the appropriate sample size for a specific investigation. • Distinguish between parameter and statistic, and probability and nonprobability sampling techniques.

Third Term	
Content Standard	The learners should have knowledge and understanding of... • compound interest; simple and general annuities; and business and consumer loans • basic concepts of: a. statistical hypothesis testing b. test of difference (parametric or non-parametric) c. test of relationships d. simple linear regression • logical propositions, syllogisms, and fallacies
Performance Standard	By the end of the term, the learners are able to... • Develop and present a digital financial proposal comparing investments with data, graphs, and evidence-based recommendations. • Design, conduct, and present a data investigation applying statistical tests and visualizations to real-life datasets. • Apply logical propositions and fallacies, demonstrating truth values and reasoning through tables, charts, and examples.
Week	Learning Competency
1	Number and Algebra • Solve problems involving compound interest, including maturity value, future value, nominal rates, and number of terms.
2	Number and Algebra • Illustrate simple annuity, general annuity, and deferred annuity. • Solve problems involving annuities.
3	Number and Algebra

	Calculate the fair market value of a cash flow stream that includes an annuity.
4	Number and Algebra - Differentiate between business and consumer loans and their characteristics, such as repayment terms, requirements, purpose, and interest rates. - Solve problems involving business and consumer loans (e.g., amortization, mortgage).
5 to 6	Data and Probability - Illustrate: - null hypothesis - alternative hypothesis - level of significance - types of error - Formulate the null and alternative hypotheses of a given problem. - Determine the: - level of significance - types of error appropriate to a given hypothesis - Construct a scatter plot and estimate the strength of association between variables. - Draw the line of best fit for a scatter plot.
6	Data and Probability - Test hypotheses for one and two population means using technology employing the following statistical tools: - z-test for known standard deviation - z-test for unknown standard deviation - one-sample t-test - independent samples t-test - paired samples t-test
7	Data and Probability Construct a scatter plot and estimate the strength of association between variables.

	• Draw the line of best fit for a scatter plot. • Determine the degree of relationship between two variables using Pearson's r correlation. • Conclude from the results of hypothesis testing on: a. one and two population means using t-test and z-test b. relationship using Pearson r correlations
8	Number and Algebra • Identify logical propositions. • Differentiate simple from compound propositions.
9	Number and Algebra • Illustrate basic logical operations. • Illustrate the different forms of conditional propositions.
10	Number and Algebra • Illustrate different types of tautologies and contradictions using practical situations. • Determine the truth values of propositions in a practical situation.